DATA-DRIVEN GEOGRAPHIC AND SOCIOECONOMIC DETERMINANTS OF FAST-FOOD DENSITY AND PUBLIC HEALTH OUTCOMES IN VIRGINIA

by

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Bachelor of Science
My Other Former School, Year of first degree

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Fairfax, VA

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Dedication

To Dr. Miller

Thank you for your mentorship and insightful guidance.

Acknowledgments

I acknowledge the data resources provided by County Health Rankings and the U.S. Census Bureau.

Also, thanks to my peers in the CDS program for their support and advice.

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Abstract

DATA-DRIVEN GEOGRAPHIC AND SOCIOECONOMIC DETERMINANTS OF FAST-FOOD DENSITY AND PUBLIC HEALTH OUTCOMES IN VIRGINIA

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George Mason University, 2025

Capstone Project Director: Dr. Kent Miller

The proposed capstone project examines how fast food restaurant density, socioeconomic characteristics, and public health outcomes in Virginia's counties are interrelated through multi-source data (OpenStreetMap, American Community Survey, and County Health Rankings 2023). This project will incorporate economic and health indicators along with geospatially based data to identify potential structural inequalities within the food environment. The density of fast food restaurants (measured per 10,000 people) for each Virginia County was calculated and compared to other socioeconomic indicators such as median household income, the degree of income disparity as expressed by the Gini Index, and the percentage of people 18 or older who are obese. ANOVA, correlation and linear regression techniques were employed as statistical methods to determine whether the characteristics of a county's economy and health indicators can help predict the degree of fast food density relative to other parts of the county or whether fast food density and poor health indicators correlate.

The research demonstrated that there is not a statistically significant association between fast-food restaurants nearby and median county income. Nevertheless, the level of

inequality of household income has demonstrated a minor yet statistically significant relationship to fast-food restaurant density. In other words, counties with more income disparity between their residents tend to be located closer together than those with less disparity. Additionally, the density of fast-food restaurants alone did not provide evidence as an independent community-level predictor of obesity rates in the adult population in Virginia; therefore, we can reason that a community's health is likely influenced by a greater range of social, environmental and health-related factors in addition to the availability of fast-food restaurants, which correlates with the level of income disparity in that area. This research supports the position that the structural socio-economic inequalities in a community shape the nature of the commercial food environment and serves as an important foundation for future consideration of policy initiatives aimed at improving nutritional equity and health in Virginia.

Chapter 1: Introduction

1.1 Background

An Analysis of the Correlation Between Fast Food Restaurant Density, Economic Status, and Health Effects in Virginia's Counties.

This research analyzes the linkages between County level adult obesity, County level Income, County level Income Inequality, and Fast Food Density in Virginia. The researcher will use data from multiple sources (including the USDA Food Environment Atlas) as well as data collected from the Virginia Department of Health, and the CDC to study the relationship between county-level economic status and adult obesity rates, and to find out how socioeconomic differences affect the local food environment and result in health disparities. It will also focus specifically on the relationship between income inequality and access to healthful foods. The study will analyze the spatial clustering of fast food restaurants, food deserts, and food swamps in the state of Virginia, looking at both County-level aggregation and county-level disaggregation data.

1.2 Main Problem

The purpose of this study is to investigate associations between socio-economic conditions and health in the Commonwealth of Virginia. In particular, we are interested in associations between fast food restaurant density, county income levels, income inequalities, and health outcomes.

1.3 Significance

Understanding connections among fast-food proximity, socio-economic measurements, and community health outcomes can serve to alleviate chronic health disparities in Virginia. Fast-food density has been widely characterized as the environmental force that determines dietary quality and obesity. Specifically, among the social communities with limited access to healthier food options were affected deeply.

This study is relevant because it shows that the Gini indicator has a statistically significant relationship using income inequality as an independent variable for fast-food outlet density in counties across Virginia. The research suggests that there are structural socioeconomic inequalities that influence the commercial food environment, and therefore a differential distribution of health risks. The findings are consistent with and add to the existing research on food deserts, and health inequities.

Furthermore, Virginia has a substantial amount of regional variation in income, rates of obesity, and food access which makes it a relevant case study in order to better understand localized public health issues. By layering geospatial data from OpenStreetMap with socioeconomic indicators from the American Community Survey (ACS) and health outcomes from the County Health Rankings dataset, this study is addressing an important gap in the literature. The use of multi-source data will also support the evidence base for policy mechanisms to reduce dietary-related disease burden or socioeconomic inequality.

In conclusion, the findings expand the understanding of structural inequality in designing public health initiatives and add important information for policymakers looking at health equity on a county level.

Chapter 2: Prior Research

2.1 Food Environment and Health

The term "food environment" has variety concepts. It refers to the accessibility, availability, and average price of food in a social community. This area of study is important for understanding obesity rate, because it is closely connected to eating habits and health outcomes. In a study from 2014, food option profiles were investigated in neighborhoods. The researchers found that in neighborhoods where people lived further from the supermarkets and the costs were higher for healthy foods like fresh vegetables and fruits, people in those neighborhoods were more overweight [1]. These neighborhoods have poor chance to access healthful food, were identified as "food deserts" in urban settings and showed a greater prevalence of obesity than others, especially in lower-income families. The researchers revisited the importance of environment and structural discourse and strayed away from comments about personal responsibility known as "willpower" to highlight the structures and environment that limited choices around nutrition and affected lifestyle disparities.

This systematic review included these primary findings and demonstrates that neighborhoods with a high density of fast food restaurants and only limited access to healthy options consistently increased the risk of obesity [2]. Specifically, "obesogenic" environments are marked by the overabundance of unhealthy food and limited physical activity opportunities and exist across various contexts, cultures, and countries.

Children in particular are impacted by their food environment significantly. Living in neighborhoods with high food density or attending schools near clustered fast food outlets is correlated with an increased incidence of childhood obesity [3]. This issue will continue to expand beyond obesity to other diseases. Food swamps refer to people living in high-density residential neighborhoods with plenty of fast food restaurants, which have been associated

with all-cause mortality and obesity-related cancers [4]. To warp up, this body of literature illustrates how the importance of the food environment is essential to understanding dietary behavior and health over the long term.

2.2 Fast-Food Density, Accessibility, and Health Outcomes

There is a considerable relationship between neighborhood density of fast-food restaurants and their proximity to where people reside. According to the study, particular higher density and proximity to fast-food restaurants were associated with a slightly higher prevalence of obesity and diabetes especially for those living in areas where there were limited parks and playgrounds [5]. These studies use average measurements like fast-food restaurant counts per square kilometer or miles and people's self-reported driving or walking time to these locations. Other research has found that even a small increase in the availability of unhealthy food can lead to an significant increasing range in community average body mass index (BMI) levels [6]. The density and accessibility of fast food restaurants are important factors how to understand health outcomes and contribute to disparities.

Moreover, this study has also found significant correlation between the density of fast food restaurants and their proximity. It provides that there is a clear inequality in community. This shows a clear inequality in the community as the minority and low income families were most likely exposed to these areas [5], [6]. These findings suggest limited accessibility of safe parks and playground spaces where people can do exercise activities, in combination with the high density and proximity to fast food. Additionally, as mentioned earlier, it ultimately affects the health of these communities disproportionately.

2.3 Rural–Urban and Socioeconomic Disparities

Research indicates that the prevalence of adolescent overweight and obesity is greater in rural, than urban settings in the U.S. [7]. Distance to the grocery stores, perceived insufficient transportation to healthy foods, and fewer youth physical activity facilities, are all shown to be associated with obesity in rural adolescents [7]. These observations have been documented in state and national policies, which provide useful evidence. For example, low income households are more likely to experience food insecurity, which has been related to obesity in a report by the Food Research & Action Center [8]. The U.S. Census Bureau has income and economic data that indicates a substantial disparity in the median income of households at the county level [9] and that these income disparities are related on diet and nutrition and other health outcomes. The Food Environment Atlas dataset from the U.S. Department of Agriculture combines food environment, health outcome, and socioeconomic indicators and provides for analyses at the county level [10].

2.4 National and Global Health Statistics

Reports from the Centers for Disease Control and Prevention/National Center for Health Statistics indicate that fast food and food from fast food restaurants are consumed by adults in all ages in the United States, especially more often consumed by younger adults, men, disabled adults, and adults with lower educational attainment [11], [12]. These high rates of fast food consumption are related to the significant prevalence of diet related diseases.

Integrating findings from other global studies, the World Health Organization noncommunicable disease profiles for the United States and other countries have obesity as a common risk factor [13]. The World Bank similarly aggregates a variety of health and income indicators which show that health disparities, including dietary and nutrition outcomes, are complex issues related to income inequality [14]. So unhealthy food environments are a

public health issue countries are experiencing around the world.

2.5 Virginia-Specific Research and Policy Reports

Focusing on Virginia tells a similar story yet adds context at the state-level. The Virginia Food Access Investment Fund annual report has state-level data on the investment made to expand healthy food retail in underserved areas [15]. Southwest Virginia, has documented food security problems where remoteness, poverty, and lack of competitive retail combine [16]

The Virginia Department of Behavioral Health and Developmental Services (DBHDS) agencies have identified nutrition vulnerabilities among at-risk populations [17], and the Virginia Rural Health Plan has identified food security as an important rural health issue [18]. As it relates to the spatial component, the Virginia Association of Counties [19] includes county boundary files and other governance related information that may be useful in accessing/using county-level data. SCHEV has state-level education data that captures sharp county-level differences in college degree attainment [20]. Taking into account the established association between education, income, and health, gaps like this help explain nutrition and obesity-related differences.

2.6 Federal Research and Policy Reports

In addition to state programs, federal programs are directly addressing food environmental inequities and health-optimizing communities as well. One prominent example of this work is the Environmental Quality Incentives Program (EQIP) High Tunnel Initiative of the Natural Resources Conservation Service (NRCS) of the United States Department of Agriculture (USDA). This program provides farmers with financial and technical assistance for the development of high tunnels and described as structures which help farmers extend their

growing season. Furthermore, assist with protection from extreme weather variabilities, and improve soil quality [21].

High Tunnel Initiative example intentionally addresses barriers to being able to access fresh and affordable produce, especially for rural and undeserved communities without supermarkets. By improving local agriculture production it improves access to fruits and vegetables in place of ultra-processed food. High tunnels further provide abilities to mitigate climate variabilities to enhance crop yield, thereby reducing price variabilities associated with fresh produce.

The High Tunnel Initiative positively expands upon Virginia's suite of state-level policy programs, such as Virginia's Food Access Investment Fund, to support a multi-scalared policy response. State level policy programs are largely limited to retail access or economic development, while Environmental Quality Incentives Program (EQIP) supports the ability of the agricultural sector to deploy prior to retail. Collectively, this initiative provides an example of the collaboration of local, state, and federal governments in multiple ways to promote food insecurity and nutrition equity.

2.7 Integration and Key Insights

In light of this, there are a few significant insights that need to emphasize to mention.

The core are accessibility and cost matter. Healthy eating options are limited or costly, and it continues to ascend obesity rates, regardless of age, geography, and culture [1], [2].

Location matters and contributes to inequities. Racially-segregated communities, lack of investment in neighborhoods and rural isolation, all of which create barriers to healthy choices and safe venues for being active, lower rates of healthy outcomes [22], [7]

Food insecurity is cyclical and endless like Mobius strip. Tight budgets, coupled and unreliable access to food, force households to ultra-processed cheap calories and ironically increases their risk for obesity, while struggling with sustained access to food [23] To cut off the strip, policy is possible, but must be location specific. Programmatic and investment strategies via healthy food retail programs (for example, Virginia's healthy-food retail grant)

could be positive first steps, but will require some form of long-term sustained effort linking tourism and or programming of this sort will need to include a place-based, large-scale agency to accomplish policy change [15].

This is a global problem. Regardless where you live, as long as unhealthy food is less expensive, and easy or far easier to get than healthy food, the variables will matter on national and global concerns.

2.8 Behavioral Responses and Consumer Psychology

Even though spatial and economic aspects of food environments have been well studied, mental and behavioral considerations are emerging as arguably the most important determinants of food choice. Behavioral economics research shows that individuals tend to have present-biased preferences, preferring to have food now, rather than weighing the later-health outcomes of the decisions they make today. This "time discounting" suggests a preference for easily accessibly and calorie dense foods is stronger in lower-income people who experience chronic stress, or at least a diminished decision bandwidth.[24]

Building on this, exposure to targeted marketing and information overload of fast-food commercials also shapes consumer attitudes and perceptions of "normal" eating behaviors.[25] With experiments and surveys, children and adolescents have reported craving and purchasing more excessive-calorie, nutrient-poor food, as a result of constant advertisement exposure, therefore educating and self-regulating aren't enough. In sum, behavioral vulnerability coupled with logistical availability works creates a stack effect that increases instances of obesogenic behavior. Given these considerations, developing behavioral concepts into the public health nutrition policy will help define food environment concerns in an integrated context.

2.9 Data-Driven Approaches and Predictive Modeling

The recent advances in computational and data sciences have greatly aided researchers in evaluating food environments and predicting health outcomes. Researchers have engaged spatial machine-learning models such as random forest regression and geographically weighted regression to assess each food environment's most powerful predictors of obesity and food insecure counties [26]. Fine grain, predictive obesity prevalence maps can be produced with combined datasets from the CDC, USDA Food Environment Atlas, and the American Community Survey to visualize regional clustering patterns of obesity prevalence.

These methods can also be used to simulate policy interventions. For example, increasing grocery store density, can be estimated to reduce county-level obesity prevalence slightly in some rural areas [27]. Machine-learning based clustering has also been used to categorize counties based on total socioeconomic and health characteristics for creating a quantifiable framework for prioritizing policy action. This advanced analytic three-prong approach captures the potential of data-driven public health completely and meshes well with the computational aspects of this capstone project.

2.10 Evaluating the Effectiveness of Food Access Programs

Many state and federal initiatives exist to increase access to food, but fewer have been rigorously evaluated. The Healthy Food Financing Initiative (HFFI), a collaborative effort of the U.S. Department of Treasury and USDA and HHS, started in 2011 to increase healthy retail in underserved communities. Longitudinal evaluations point to improvement in the presence of retail, but with limited effect in changing BMI or fruit-and-vegetable consumption without nutrition education and community engagement [28].

Similarly, the CDC's State Indicator Report on Fruits and Vegetables (2022) expresses that States that offer nutrition education, have incentives for local agriculture, and take an urban planning approach tend to show greater improvements in dietary behavior [29]. The Robert Wood Johnson Foundation, in a Healthy Communities Initiative, also finds

that programs that are successful often couple increasing access to physical food with high-level policy support such as zoning reform and improving school meals [30]. Evaluation approaches cite that access is not sufficient for longer-term impact, and multi-sector improvement and coordination are needed.

2.11 Digital Literacy and Health Promotion Campaigns

In the present digital age, nutrition knowledge and behavior change are increasingly delivered via online and mobile app spaces. For example, Fewer examples of public health campaigns that have used social media include the NIH "MyPlate on the Go" and CDC "Fruits and Veggies–More Matters" programs, both of which have demonstrated measurable change in dietary knowledge and self-reported healthy eating among young adults [30].

Digital interventions further help to address access to nutrition, health, and behavioral health information in rural and low-income communities by providing culturally-appropriate public health nutrition content and helping to increase participation in virtual community health challenges. However, misinformation surrounding current diet trends on digital platforms like TikTok and YouTube can complicate nutritional literacy [31]. Therefore, the potential benefit of integrating evidence-based digital health education, in conjunction with local policy interventions, and ultimately change practitioners of healthy eating can improve overall capacity infrastructure for healthy eating.

Chapter 3: Data

3.1 Fast-Food Restaurant Data

Specific fast-food restaurant location data were retrieved using by OpenStreetMap (OSM) Overpass API. Queries filtered for establishments tagged as `amenity=fast_food` across all counties and independent cities in Virginia. Locations classified as amenity=fast food were filtered through queries that encompassed all counties and independent cities in Virginia. The data included latitude, longitude, and business name. For aggregation and density by county, latitude and longitude were geospatially joined with county-level geometries constructed from the U.S. Census Bureau's TIGER/Line shapefiles.

3.2 Population and Income Data

Demographic data were gathered for total population and median household income by county through the `tidycensus` R package utilizing the U.S. Census Bureau's 2023 American Community Survey (ACS) 5-Year Estimates. FIPS states were added to justify that the data were spatially aligned for analysis.

3.3 Health Data

County-level health outcomes were sourced from the 2023 County Health Rankings & Roadmaps dataset, provided by the University of Wisconsin Population Health Institute and the Robert Wood Johnson Foundation. The variable of interest was Adult Obesity, defined as the percentage of adults (aged 20 and older) with a body mass index (BMI) of 30 or higher. The dataset was filtered to retain Virginia counties only, and joined to the primary dataset by county name and FIPS code.

3.4 Data Preprocessing and Analysis Workflow

3.4.1 Geospatial Data Cleaning

The fast-food restaurant dataset was obtained from OpenStreetMap using the `amenity=fast_food` tag for all locations in Virginia. Using the `sf` package, the data was loaded in GeoJSON format. If geometries were not of type `POINT`, they were converted using `st_point_on_surface()` to ensure spatial compatibility. Entries without valid business names were removed, and duplicates were filtered by OpenStreetMap ID where available.

3.4.2 County Assignment and Density Calculation

Using TIGER/Line shapefiles accessed via the `tigris` package, county boundaries for Virginia were imported. A spatial join (`st_join`) was performed to assign each restaurant to its containing county. The number of fast-food restaurants per county was then counted, and population data from the ACS 2023 dataset (`B01003_001`) was joined via the `tidycensus` API. Fast-food density per 10,000 residents was calculated as:

$$\text{Fast-Food Density} = \frac{\text{Number of Restaurants}}{\text{Total Population}} \times 10,000$$

3.4.3 Socioeconomic Variable Integration

Median household income (`B19013_001`), Gini index (`B19083_001`), and income quintile ratios (`B19081`) were extracted from the 2023 ACS 5-Year Estimates. After reshaping with `pivot_wider()`, a new variable was calculated as the ratio between top and bottom income quintiles:

$$\text{Income Ratio} = \frac{\text{Top 20\% Income}}{\text{Bottom 20\% Income}}$$

These variables were joined with the fast-food density dataset using GEOID as the primary key.

3.4.4 Health Data Integration

Adult obesity rates by county were extracted from the *2023 County Health Rankings & Roadmaps* Excel file. The FIPS codes were standardized and matched with county GEOIDs to ensure alignment. The obesity rate was extracted as the percentage of adults with a $BMI \geq 30$.

3.4.5 Statistical Analysis

Counties were categorized into tertiles (Low, Mid, High) based on median household income using `ntile()` from the `dplyr` package. An ANOVA test was conducted to assess the differences in fast-food density across income groups:

$$H_0 : \mu_{\text{Low}} = \mu_{\text{Mid}} = \mu_{\text{High}} H_A : \text{At least one group differs}$$

Tukey's HSD post-hoc test was used to identify which income groups differed.

3.4.6 Regression Models

Two linear regression models were built:

1. Fast-food density predicted by income:

$$\texttt{fastfood_per10k} \sim \texttt{median_income}$$

2. Fast-food density predicted by all socioeconomic factors:

$$\texttt{fastfood_per10k} \sim \texttt{median_income} + \texttt{gini} + \texttt{income_ratio}$$

Finally, a regression model was also used to estimate the relationship between fast-food density and adult obesity rate:

$$\texttt{obesity_rate} \sim \texttt{fastfood_per10k}$$

3.4.7 Visualization

All maps were produced using `ggplot2` and `viridis` color scales. County-level choropleths visualized:

- Raw fast-food counts
- Fast-food density per 10,000 residents
- Median household income
- Gini index
- Side-by-side comparison of density and income using `cowplot`

All processing, analysis, and visualization were conducted entirely within the R environment.

Chapter 4: Theory

4.1 Research Hypothesis

This study investigates the relationships among county-level income, income inequality, fast-food access, and public health outcomes in the state of Virginia. The following hypotheses are formulated based on existing literature on food environments, spatial inequality, and social determinants of health.

4.1.1 Hypothesis Set 1: Median Income and Fast-Food Density

- **Null Hypothesis ($\mathbf{H}_{01}$):** There is no statistically significant association between median household income and fast-food density across counties in Virginia.
- **Alternative Hypothesis ($\mathbf{H}_a$):** Counties with lower median household income exhibit higher fast-food density.

4.1.2 Hypothesis Set 2: Income Inequality and Fast-Food Density

- **Null Hypothesis ($\mathbf{H}_{02}$):** Income inequality (Gini index) is not significantly associated with fast-food density.
- **Alternative Hypothesis ($\mathbf{H}_a$):** Counties with higher income inequality exhibit higher fast-food density.

4.1.3 Hypothesis Set 3: Public Health Outcomes and Fast-Food Density

- **Null Hypothesis ($\mathbf{H}_{05}$):** Fast-food density is not significantly associated with obesity rates.

- **Alternative Hypothesis (H_a):** Counties with higher fast-food density have higher obesity rates

4.1.4 Potential Statistical Tests

Based on the final dataset, the following statistical models may be applied:

- Pearson correlation test
- Ordinary Least Squares (OLS) regression
- ANOVA model, when comparing groups of counties categorized by income level
- Geographically Weighted Regression (GWR) for spatial heterogeneity (optional)

An example ANOVA model formulation:

$$\text{FastFoodDensity}_i = \mu + \alpha_j + \varepsilon_i,$$

where α_j represents the effect of county income group j (e.g., high-income vs. low-income), and ε_i is random error.

4.2 Socioeconomic Status

To show the disparity in access to fast food across the state of Virginia, several choropleth maps were produced by conducting spatial joins with OpenStreetMap (OSM) fast food location data and the 2023 U.S. Census ACS county boundaries.

The top map shows the raw number of fast food restaurants by county. Notice that cities (represented as counties) such as Fairfax and Virginia Beach have a high concentration of fast food units, which suggests that commercial food units cluster in areas with population density (urban areas).

To represent density by population levels, fast food restaurant density per 10,000 residents was calculated. This measure better represents relative access to fast food, especially

for rural counties that have relatively high density per capita (e.g., Emporia, Galax), but will have fewer raw units because of their lower overall population.

County income distributions serve as useful socioeconomic context. Northern Virginia counties have high median incomes, but the majority of southern and southwestern counties have low household income. The geographic disparity of income distribution will be the basis of subsequent analysis of fast food units by income group.

Interpretation of Geospatial Visualizations

Maps of spatial distribution indicate that counties in southwestern and southeastern Virginia have lower median incomes and higher fast-food density. The adult obesity map exhibits similar spatial patterns, with higher rates of obesity in areas of high fast-food density and income inequality, thus supporting the regression analysis results.

Such geographic distributions present visual support for the statistical relationship between socio-economic variables and health-related outcomes.

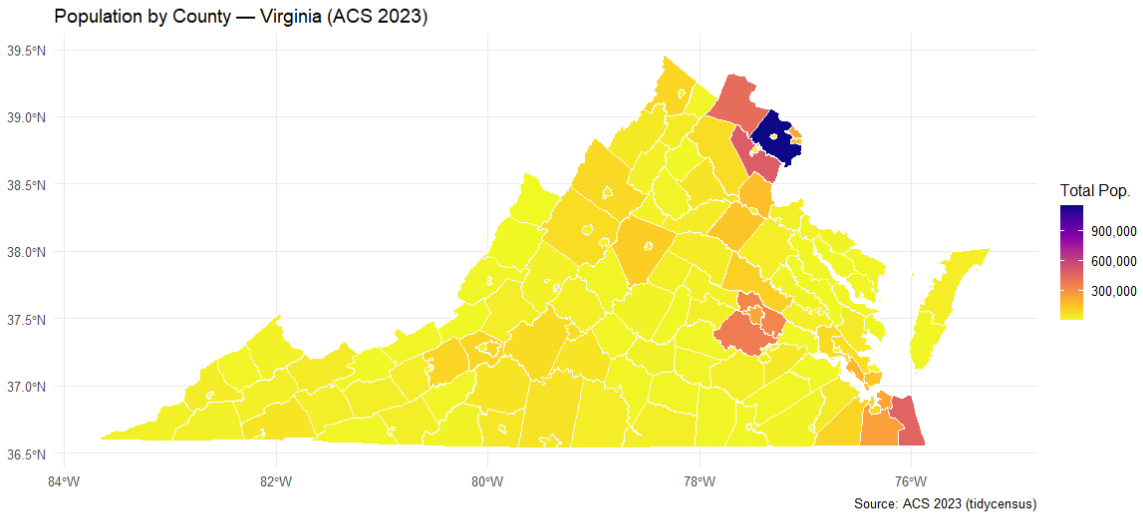


Figure 4.1: Population Status in Virginia

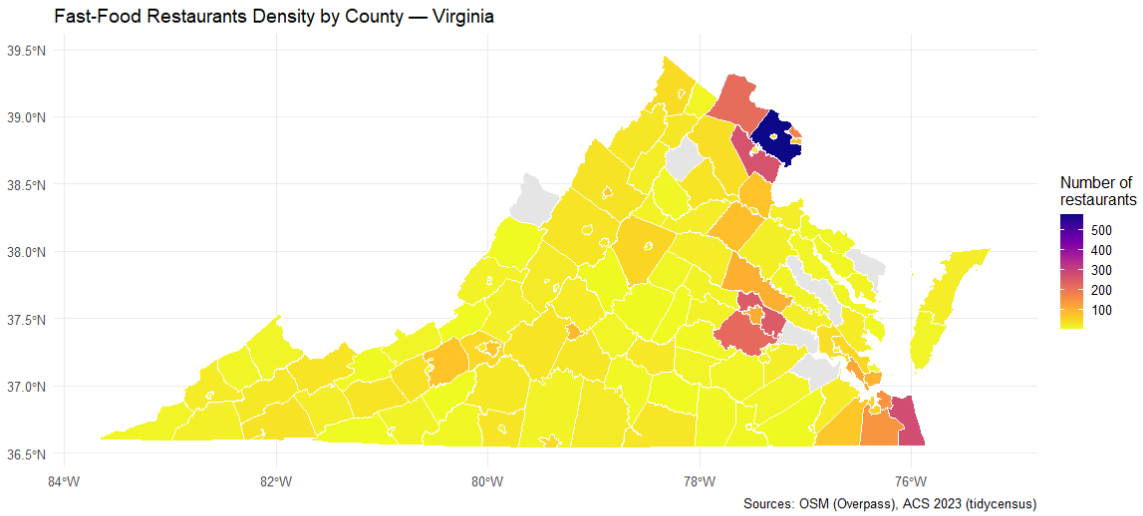


Figure 4.2: Fast Food Density in Virginia

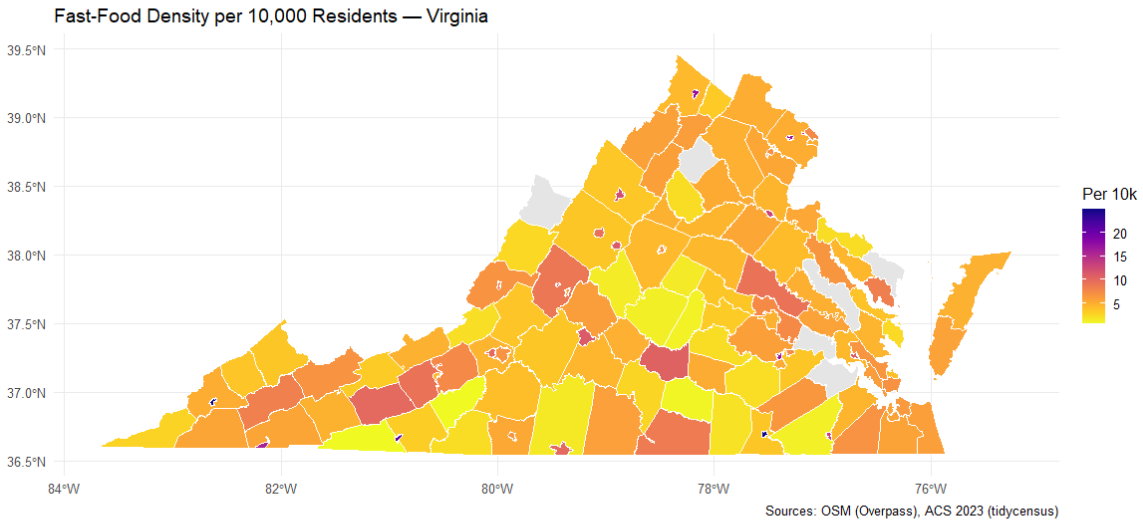


Figure 4.3: Fast Food Density per Residents(10K) in Virginia

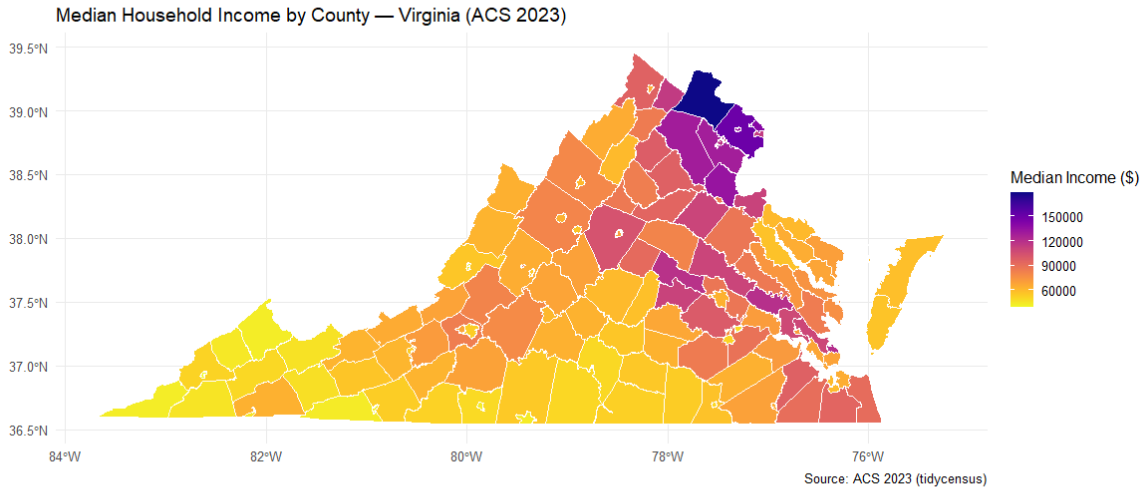


Figure 4.4: Median Income map in Virginia

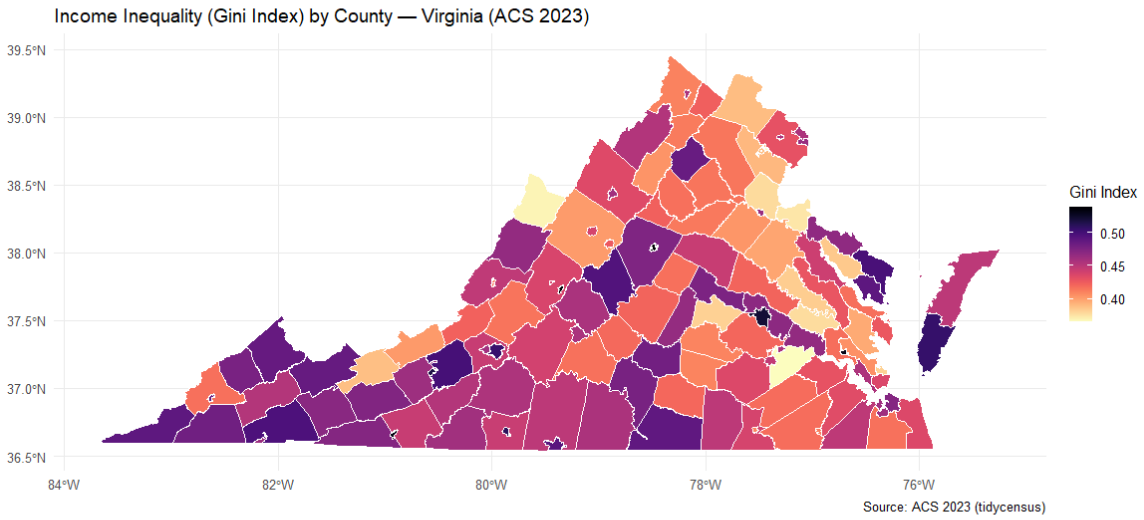


Figure 4.5: Income Inequality(Gini Index) in Virginia

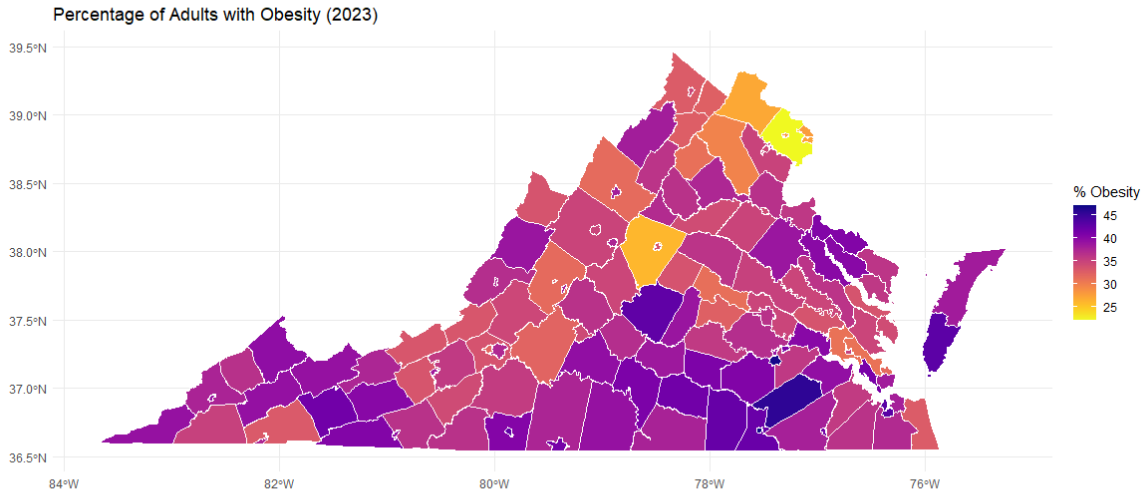


Figure 4.6: Percentage of Adults with Obesity

Chapter 5: Results

This study seeks to understand the relationship between the geographic distribution of fast food establishments and socioeconomic or health-related factors at the county level in Virginia. The underlying theoretical assumption is that higher fast food density may reflect and influence both local economic disparities and public health outcomes.

As shown in Table 5.1, both tests yielded p-values greater than 0.05. We therefore fail to reject the null hypothesis and conclude that there is no statistically significant relationship between fast food restaurant density and median income levels across counties in Virginia.

5.1 Regression Models

As an initial test, we conducted both an ANOVA and a linear regression to examine whether there is a statistically significant association between fast food density (number of fast food restaurants per 10,000 residents) and median household income by county.

Table 5.1: p-value of ANOVA and Linear Regression: Fast Food Density by Median Income

Test	p**-value**
ANOVA (Income vs Fast Food Density Groups)	0.653
Linear Regression (y = Density, x = Median Income)	0.146

As shown in Table 5.1, both the ANOVA and linear regression results yield p-values greater than 0.05, which suggests that there is no statistically significant association between median household income and fast food restaurant density. Therefore, the null hypothesis cannot be rejected at the $\alpha = 0.05$ level.

Table 5.2: ANOVA Summary: Fast Food Density by Income Group

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Income Group	2	36.2	18.10	0.957	0.387
Residuals	124	2345.3	18.91		

Table 5.3: Linear Regression Summary: Fast Food Density vs. Median Income

Predictor	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.151e+00	1.129e+00	6.334	< 0.001 ***
Median Income	-1.246e-05	1.415e-05	-0.881	0.380

Residual standard error: 4.351 on 125 degrees of freedom
Multiple R^2: 0.006171, Adjusted R^2: -0.00178
F-statistic: 0.7761 on 1 and 125 DF, p-value: 0.38

Fast Food Density vs. Income Analysis

To examine the relationship between fast food outlet density and income levels across counties in Virginia, we conducted both a one-way ANOVA and a simple linear regression using the variable `fastfood_per10k` (number of fast food outlets per 10,000 residents).

The ANOVA result shows no significant difference in fast food density across income groups ($p = 0.387$). Post-hoc analysis using Tukey HSD confirms this result, with all pairwise p-values > 0.40.

These findings indicate that median income is not a statistically significant predictor of fast food density in Virginia counties. The regression model explains a negligible amount of variance ($R^2 \approx 0.006$), and the slope coefficient is not statistically significant ($p = 0.38$). Thus, we find no strong evidence of income-based disparity in fast food access at the county level.

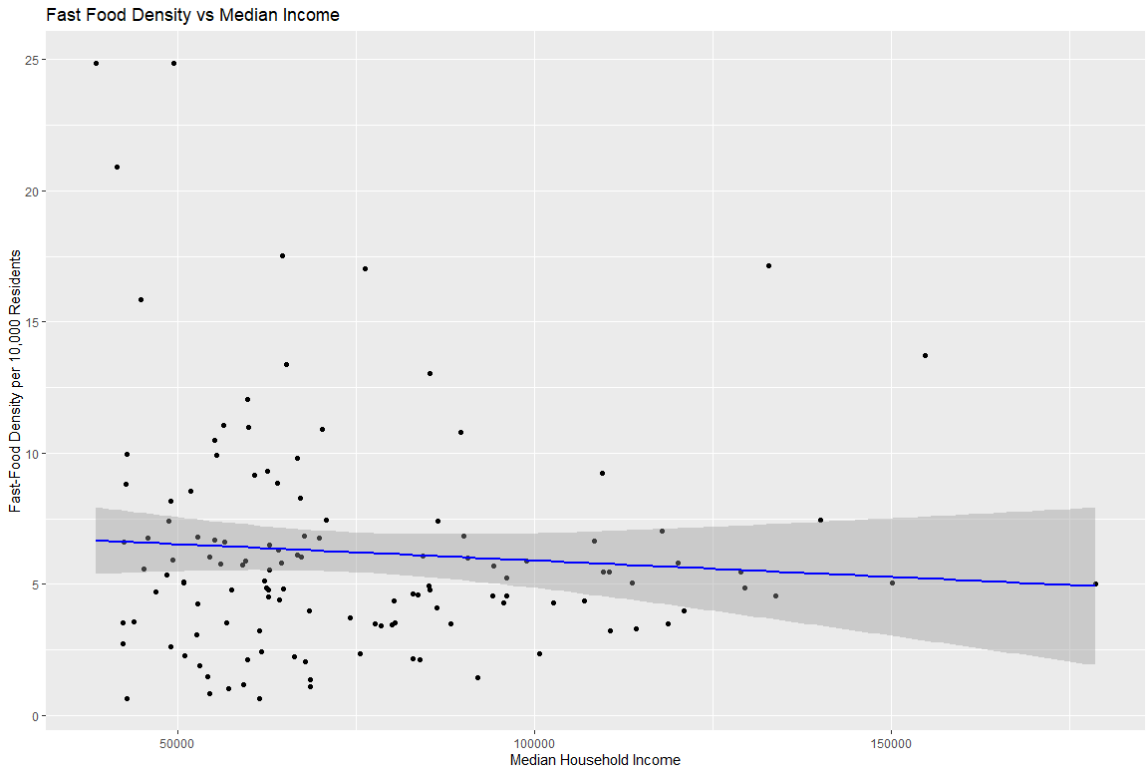


Figure 5.1: Fast Food Density vs. Median Income in Virginia

Fast-Food Density and Median Household Income

To explore the relationship between income level and fast-food availability, we conducted a simple linear regression with fast-food density (restaurants per 10,000 residents) as the dependent variable and median household income as the independent variable.

Regression Results The estimated regression model is:

$$\widehat{\text{Fast-Food Density}} = 7.15 - 0.0000125 \times \text{Median Income}$$

The slope coefficient was not statistically significant ($\beta_1 = -0.0000125$, $p = 0.38$), indicating that median income was not a meaningful predictor of fast-food density at the county level. The model explains less than 1% of the variance in fast-food density ($R^2 = 0.0062$), and the F-test for the model was not significant ($F(1, 125) = 0.7761$, $p = 0.38$).

Table 5.4: Simple Linear Regression: Fast-Food Density on Median Household Income

Variable	**Estimate**	**Std. Error**	**t value**	**p value**
(Intercept)	7.151	1.129	6.334	<0.001
Median Income	-1.246×10^{-5}	1.415×10^{-5}	-0.881	0.38
Residual Std. Error: 4.351 on 125 DF				
Multiple R^2: 0.0062, Adjusted R^2: -0.0018				
F-statistic: 0.7761 on 1 and 125 DF, p-value: 0.38				

Findings Contrary to popular assumptions, the regression results suggest that there is no statistically significant linear relationship between median household income and fast-food density across Virginia counties. This supports the notion that fast-food availability may be influenced by other factors, such as zoning, urbanicity, or demographic composition.

Income Inequality (Gini Index) by Fast-Food Density

To examine the association between income inequality and the density of fast-food restaurants across counties in Virginia, we conducted both a correlation analysis and a linear regression using the Gini Index as the predictor.

Pearson Correlation The Pearson correlation coefficient between fast-food density (per 10,000 residents) and the Gini Index was found to be 0.2095, indicating a weak but positive correlation. A simple linear regression yielded the following model:

$$\text{FastFood_per10k} = -4.377 + 23.939 \cdot \text{Gini}$$

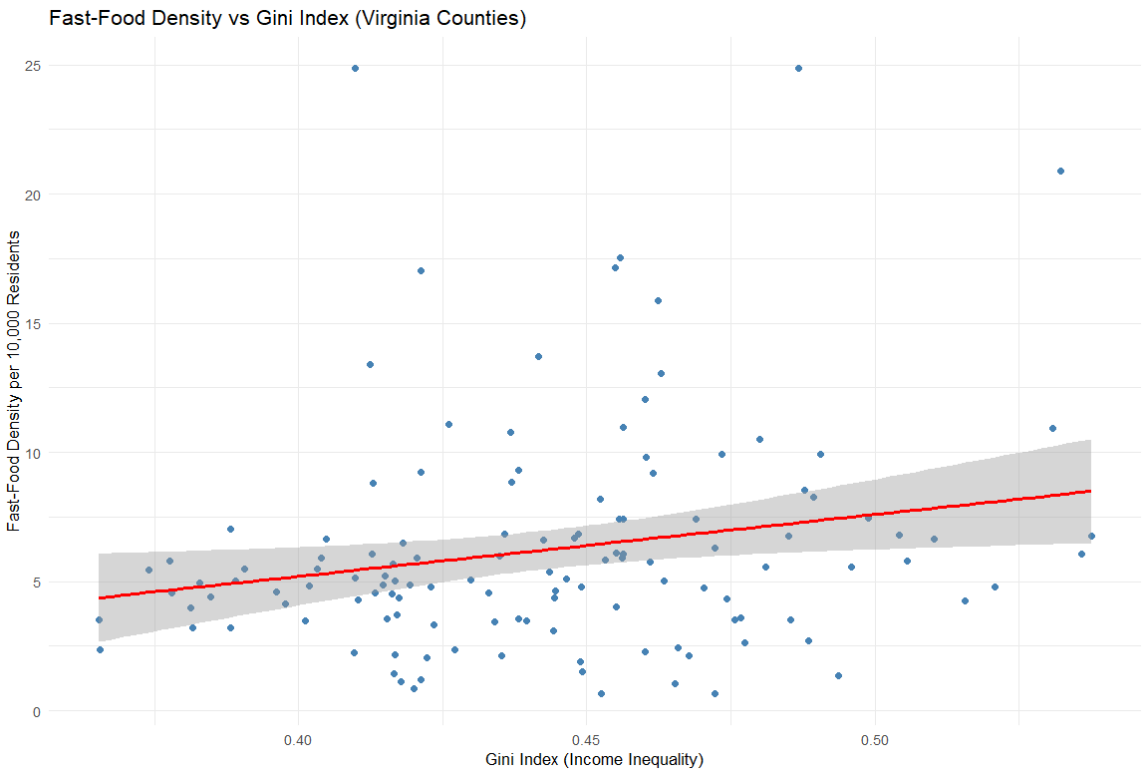


Figure 5.2: Gini Index in Virginia

Table 5.5: Simple Linear Regression: Fast-Food Density on Gini Index

Variable	**Estimate**	**Std. Error**	**t value**	**p value**
(Intercept)	-4.377	4.439	-0.986	0.3260
Gini Index	23.939	9.994	2.395	0.0181
Residual Std. Error: 4.295 on 125 DF				
Multiple R^2: 0.0439, Adjusted R^2: 0.0362				
F-statistic: 5.734 on 1 and 125 DF, p-value: 0.0181				

Findings The coefficient for the Gini Index was statistically significant (p = 0.018), suggesting that higher income inequality is associated with greater fast-food restaurant density. The model's explanatory power was limited, with an R^2 of 0.044, indicating that income inequality explains about 4.4% of the variation in fast-food density.

Fast-Food Density and Obesity rates

While our statistical analysis did not find a significant relationship between fast food density and income level, it remains important to consider how the spatial concentration of fast food establishments may affect public health outcomes directly.

Food Desert Theory posits that low-income communities often experience limited access to healthy food options, leading to a reliance on calorie-dense, nutrient-poor alternatives such as fast food [32].

Prior research has identified correlations between higher fast food availability and elevated obesity rates and diet-related chronic conditions such as type 2 diabetes, particularly in underserved or marginalized populations [33, 34].

Thus, even when there is no direct relationship with income, exploring the potential health effects of fast food access is essential for assessing larger trends of health inequities in communities.

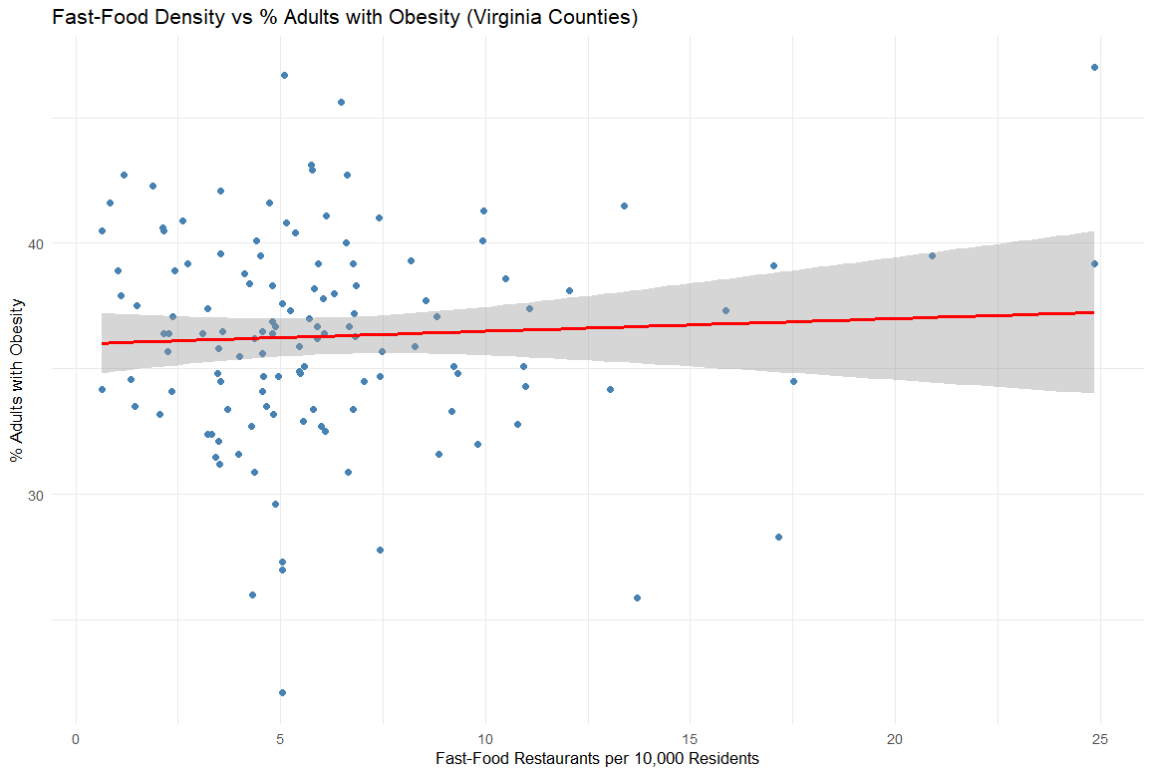


Figure 5.3: Fast Food Density vs. Adults with Obesity in Virginia

To assess our predictions that fast food density is predictive of obesity prevalence at the county level in Virginia, we ran three statistical tests: a Pearson correlation, a one-way ANOVA, and a simple linear regression. Data on the obesity prevalence was taken from the 2023 County Health Rankings, and the fast food density was operationalized as number of fast food outlets per 10,000 population.

Pearson Correlation The Pearson correlation coefficient between fast-food density and adult obesity rate was:

$$r = 0.053$$

This value indicates a very weak positive correlation, suggesting little to no linear association between the two variables. Visual inspection of the scatterplot (Figure 5.3) confirms the lack of a meaningful linear trend.

ANOVA Analysis To explore possible non-linear group differences, counties were categorized into tertiles based on fast-food density (Low, Medium, High). A one-way ANOVA was conducted to compare mean obesity rates across the three groups. The results are summarized in Table 5.6.

Source	Df	Sum Sq	Mean Sq	F-value	p-value
Between Groups	2	32.1	16.05	1.37	0.259
Within Groups	124	1454.2	11.73		

Table 5.6: ANOVA Summary: Obesity Rate by Fast-Food Density Groups

The p-value ($p = 0.259$) indicates that there is **no statistically significant difference** in average adult obesity rates across the Low, Medium, and High fast-food density groups. Therefore, we fail to reject the null hypothesis.

Regression Model Result A regression model was estimated to determine whether fast-food density significantly predicts obesity prevalence:

$$\text{Obesity Rate}_i = \beta_0 + \beta_1 \cdot \text{FastFoodDensity}_i + \varepsilon_i$$

Regression results are shown in Table 5.7.

Predictor	Estimate	Std. Error	p-value
Intercept (β_0)	30.28	0.79	< 0.001
Fast-Food Density (β_1)	0.021	0.037	0.566

Table 5.7: Linear Regression Summary: Fast-Food Density Predicting Obesity Rate

There was no statistically significant slope coefficient ($p = 0.566$), and the model showed that less than 1% of the variance in obesity ($R^2 \approx 0.003$) was explained by fast food density. Thus, density of fast-food does not seem to be a decent predictor of county-level obesity rates.

Findings The findings from the correlation, ANOVA, linear regression suggest that fast-food density alone is not a good predictor of obesity rates across Virginia's counties. The data suggest that beyond fast food density, social, environmental, and behavioral determinants play a role in obesity. Future analyses will examine several potential predictors, including income and income inequality, access to physical activity, and food insecurity to develop multivariate explanatory models.

Chapter 6: Conclusion

Overall this research study, the effect of fast food restaurant density with respect to the average income level in terms of adult obesity and county level based on 2023, was evaluated to determine whether there were any links between these three variables at the county level in Virginia. Many interesting discoveries came from this research. First, it was found that there were no significant relationships between fast-food restaurant density and median household income. This shows that fast food restaurants may be located primarily in population dense commercial centers regardless of household income levels. Second, it was determined that income inequality had a significantly positive association with fast food density. This means that the inequalities occurring in communities are much more significant than income level when determining the local food environment, which matches previous studies emphasizing that structural inequalities affect how communities obtain and use unhealthy foods.

Fast-food density does not predict the nature of the correlation multiple county comparisons in Virginia, including smoothness and linearity. Therefore, the diet-health-fast-food relationship does not depend solely on the density of a population within the state, but rather upon several other variables, including behaviors, environments, cultural influences, and the way policies impact how a person accesses and consumes unhealthy food. The availability of fast food may be perceived as a marker of obesogenic environments, but the findings of this research indicate that this single measure does not sufficiently explain the variance between populations' health levels and obesity levels across the Virginia counties studied.

Most studies show that structural inequities in an individual community's food system directly correlate with the overall health of a community. Although researchers have primarily focused on the number of fast-food outlets in a community, there may be other

areas of consideration, such as; food insecurity, physical activity access, community zoning and community culture-related behaviors, that need to be included when examining health inequities in any particular community. If policymakers wish to address health equity and improve health outcomes for rural residents, then all policymakers must work collaboratively across sectors to address multi-level inequities related to access to healthy foods and services in their communities, not just for increasing access points to food retail outlets.

Chapter 7: Software

Listing 7.1: Full R script for Fast-Food, Income, Population, and Obesity Analysis in Virginia

```
#################################################################
# 0. Load packages & options
#################################################################
library(sf)
library(tidyverse)
library(dplyr)
library(tigris)
library(tidycensus)
library(readr)
library(stringr)
library(ggplot2)
library(viridis)
library(cowplot)
library(readxl)

options(tigris_use_cache = TRUE)

#################################################################
```

```
# 1. Read & clean Fast-Food OSM data
#################################################################
ff <- st_read("fast_food_va.geojson", quiet = TRUE)

# Set CRS if missing and transform to WGS84
if (is.na(st_crs(ff))) st_crs(ff) <- 4326
ff <- st_transform(ff, 4326)

# Convert non-point geometries to point centroids
if (!all(st_geometry_type(ff) %in% c("POINT", "MULTIPOINT"))) {
  ff <- st_point_on_surface(ff)
}

# Remove duplicates by OSM ID
if ("id" %in% names(ff)) {
  ff <- ff |> distinct(id, .keep_all = TRUE)
}
nrow(ff)

# Select core columns
core_cols <- c(
  "id", "name", "brand", "amenity", "cuisine",
  "addr:city", "addr:street", "addr:housenumber"
)
```

```
ff_clean <- ff |>
  select(any_of(core_cols), geometry) |>
  filter(!is.na(name))    # Remove rows without a name

ff_attr <- ff_clean |> st_drop_geometry()

unknown_names <- ff_attr |>
  filter(is.na(brand), !is.na(name)) |>
  count(name, sort = TRUE)

#################################################################
# 2. County boundaries + fast-food counts, population, income
#################################################################

va_cnty <- counties(state = "VA", year = 2023, cb = TRUE, class = "sf") |>
  st_transform(4326) |>
  select(GEOID, NAME, geometry)

ff_join <- st_join(ff_clean, va_cnty, join = st_within) |>
  filter(!is.na(GEOID))

ff_count <- ff_join |>
  st_drop_geometry() |>
  count(GEOID, NAME, name = "fastfood_count") |>
  arrange(desc(fastfood_count))
```

```
brand_col <- intersect(c("brand", "name"), names(ff_join))[1]

ff_brand <- ff_join |>
  st_drop_geometry() |>
  mutate(chain = if (!is.null(brand_col)) .data[[brand_col]] else NA_characte
  mutate(chain = ifelse(is.na(chain), "Unknown", str_squish(chain))) |>
  count(chain, sort = TRUE)

pop <- get_acs(
  geography = "county",
  variables = "B01003_001",
  state = "VA",
  year = 2023,
  survey = "acs5",
  geometry = FALSE
) |>
  select(GEOID, population = estimate)

ff_den <- ff_count |>
  left_join(pop, by = "GEOID") |>
  mutate(fastfood_per10k = (fastfood_count / population) * 10000)

write_csv(ff_den, "va_fastfood_by_county.csv")
```

```
ff_den |> arrange(desc(fastfood_per10k)) |> head(10)


income_va <- get_acs(
  geography = "county",
  state     = "VA",
  year      = 2023,
  survey    = "acs5",
  variables = c(
    median_income = "B19013_001",
    gini          = "B19083_001",
    bottom20      = "B19081_001",
    top20         = "B19081_005"
  ),
  geometry = FALSE
)


income_va <- income_va |>
  select(GEOID, NAME, variable, estimate) |>
  tidyr::pivot_wider(
    names_from = variable,
    values_from = estimate
  ) |>
  mutate(income_ratio = top20 / bottom20)


va_final <- ff_den |>
```

```
  left_join(income_va, by = "GEOID")

##################################################################
# 3. MAPS
##################################################################

va_map <- va_cnty |> left_join(ff_den, by = c("GEOID", "NAME"))

ggplot(va_map) +
  geom_sf(aes(fill = fastfood_count), color = "white", linewidth = 0.1) +
  scale_fill_viridis(option = "C", direction = -1) +
  labs(
    title = "Fast-Food Restaurants by County — Virginia",
    fill = "Count",
    caption = "OSM + ACS 2023"
  ) +
  theme_minimal()

ggplot(va_map) +
  geom_sf(aes(fill = fastfood_per10k), color = "white", linewidth = 0.1) +
  scale_fill_viridis(option = "C", direction = -1) +
  labs(
    title = "Fast-Food Density per 10,000 Residents — Virginia",
    fill  = "Per 10k",
    caption = "OSM + ACS 2023"
```

```
  ) +
  theme_minimal()

va_income_map <- get_acs(
  geography = "county",
  state = "VA",
  year = 2023,
  survey = "acs5",
  variables = c(median_income = "B19013_001"),
  geometry = TRUE
) |>
  select(GEOID, NAME, median_income = estimate, geometry)

ggplot(va_income_map) +
  geom_sf(aes(fill = median_income), color="white", size=0.1) +
  scale_fill_viridis(option="C", direction=-1) +
  labs(title="Median Household Income (ACS 2023)", fill="Income") +
  theme_minimal()

va_gini_map <- get_acs(
  geography = "county",
  state = "VA",
  year = 2023,
  survey = "acs5",
  variables = c(gini = "B19083_001"),
```

```
  geometry = TRUE
) |>
  select(GEOID, NAME, gini = estimate, geometry)

ggplot(va_gini_map) +
  geom_sf(aes(fill = gini), color="white", size=0.1) +
  scale_fill_viridis(option="A", direction=-1) +
  labs(title="Income Inequality (Gini Index)", fill="Gini") +
  theme_minimal()

pop_va <- get_acs(
  geography = "county",
  state = "VA",
  year = 2023,
  survey = "acs5",
  variables = "B01003_001",
  geometry = FALSE
) |> select(GEOID, population = estimate)

va_pop_map <- va_cnty |> left_join(pop_va, by = "GEOID")

ggplot(va_pop_map) +
  geom_sf(aes(fill = population), color="white", size=0.1) +
  scale_fill_viridis(option="C", direction=-1, labels=scales::comma) +
  labs(title="Population by County (ACS 2023)", fill="Population") +
```

```
  theme_minimal()

health <- read_excel(
  "2023 County Health Rankings Virginia Data - v3.xlsx",
  sheet = "Ranked Measure Data",
  skip = 1
)

health <- health |> mutate(FIPS = str_pad(as.character(FIPS), 5, pad="0"))

obesity_cols <- names(health)[str_detect(names(health), "Obesity")]

health_obesity <- health |>
  select(FIPS, County, all_of(obesity_cols))

va_obesity_map <- va_cnty |>
  left_join(health_obesity, by = c("GEOID" = "FIPS"))

obesity_col_map <- names(va_obesity_map)[str_detect(names(va_obesity_map), "C

ggplot(va_obesity_map) +
  geom_sf(aes(fill = .data[[obesity_col_map]]), color="white", size=0.1) +
  scale_fill_viridis(option="C", direction=-1) +
  labs(title="% Adults with Obesity (2023)", fill="% Obesity") +
  theme_minimal()
```

```
##############################################################
# 4. ANOVA
##############################################################

va_final_anova <- va_final |>
  filter(!is.na(median_income), !is.na(fastfood_per10k)) |>
  mutate(
    income_group = ntile(median_income, 3),
    income_group = factor(income_group, labels = c("Low", "Mid", "High"))
  )

va_final_anova |>
  group_by(income_group) |>
  summarise(
    count = n(),
    mean_fastfood = mean(fastfood_per10k),
    sd_fastfood = sd(fastfood_per10k),
    .groups = "drop"
  )

anova_result <- aov(fastfood_per10k ~ income_group, data = va_final_anova)
summary(anova_result)

TukeyHSD(anova_result)
```

```
##############################################################
# 5. Regression Models
##############################################################

cor(va_final$fastfood_per10k, va_final$median_income, use="complete.obs")
cor(va_final$fastfood_per10k, va_final$gini, use="complete.obs")
cor(va_final$fastfood_per10k, va_final$income_ratio, use="complete.obs")

merged <- va_final |>
  left_join(health_obesity, by = c("GEOID" = "FIPS"))

obesity_col <- names(merged)[str_detect(names(merged), "Obesity")][1]

cor_value <- cor(merged$fastfood_per10k, merged[[obesity_col]], use="complete
print(paste("Correlation with Obesity:", round(cor_value, 3)))

model_gini <- lm(fastfood_per10k ~ gini, data = merged)
summary(model_gini)

ggplot(merged, aes(x=gini, y=fastfood_per10k)) +
  geom_point(color="steelblue") +
  geom_smooth(method="lm", color="red") +
  labs(
    title="Fast-Food Density vs Gini Index",
```

```
    x="Gini Index",
    y="Fast-Food Restaurants per 10k residents"
  ) +
  theme_minimal()

lm_income <- lm(fastfood_per10k ~ median_income, data = merged)
summary(lm_income)

ggplot(merged, aes(x=median_income, y=fastfood_per10k)) +
  geom_point() +
  geom_smooth(method="lm", color="blue") +
  labs(
    title="Fast-Food Density vs Median Income",
    x="Median Household Income",
    y="Fast-Food per 10k"
  ) +
  theme_minimal()

ggplot(merged, aes(x=fastfood_per10k, y=.data[[obesity_col]])) +
  geom_point(color="steelblue") +
  geom_smooth(method="lm", color="red") +
  labs(
    title="Fast-Food Density vs % Adults with Obesity",
    x="Fast-Food Restaurants per 10k",
    y="% Adults with Obesity"
```

```
) +
theme_minimal()
```

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